

Statement of Work
2026 Cyclic Inspection and Cleaning Water Reservoirs
Blue Ridge Parkway
Alleghany, Wilkes, Watauga, Avery, Mitchell, and Yancey Counties, NC

PART 1: GENERAL

1-1 DESCRIPTION OF WORK

- A. This contract provides services to inspect and clean underground, potable, concrete water reservoirs located on the Highlands and Pisgah Districts of the Blue Ridge Parkway (BLRI) from the Cumberland Knob Area (Alleghany County, NC @ MP 1217.3) to the Crabtree Meadows Area (Yancey County, NC @ MP 339.5). This contract will inspect all identified water reservoirs providing a detailed condition assessment and proposed mitigation plan for deficiencies discovered, as well as, a basic cleaning of the interior surfaces of each water reservoir properly disposing of the chlorinated water, sludge, debris and other waste. In-Service Cleaning and Inspection of Potable Water Storage Reservoirs may use Commercial Divers or Remotely Operated Vehicle (ROV) as the need dictates. In service cleaning shall be performed prior to tank inspection.
- B. The contract work includes:
1. Basic cleaning of each reservoir's interior surfaces.
 2. Inspection of the interior and exterior of the water reservoirs.
 3. Provide detailed inspection notes, repairs options, and inspection results in hard copy and electronically.
 4. Contract base bid and bid additives includes
 - a. Base bid - Six (6) below ground water storage tank cleaning, inspections.
 - b. Bid Additive 1 - Two (2) below ground water storage tank cleaning, inspections.
 - c. Bid Additive 2 - Two (2) above ground water storage tank cleaning, inspections.
 - d. Bid Additive 3 - Three (3) below ground water storage tank cleaning, inspection.
- C. All work will be performed under a single contract.

1-2 BACKGROUND

- A. There are numerous concrete poured in-place water reservoirs, and steel elevated storage tanks located along the Blue Ridge Parkway both in Virginia and North Carolina. The northernmost reservoir, for this project, is located at the Smart View Picnic Area (Franklin County, VA @ MP 154.5) with the other tanks located along the Parkway ending at the Blue Ridge Music Center (Carroll County, VA @ MP 212.9).

1-3 LOCATION AND ACCESS

- A. **Location:** The Plateau District is located along the Blue Ridge Parkway in Virginia. Locations of tanks are listed by site location and Mile Post (MP) along the Parkway. Reservoir sizes are included:
1. **Base Bid locations:** (tank cleaning, and inspections)
 - a. Cumberland Knob Area – 15,000 Gallon Reservoir (MP 217.3)
 - b. Bluffs Maintenance Area – 10,000 Gallon Reservoir (MP 245.5)
 - c. Jeffress Park-Cascades Area – 10,000 Gallon Reservoir (MP 271.5)
 - d. Cone Park Area – 15,000 Gallon Reservoir (MP 294.6)
 - e. Price Park Area – 60,000 Gallon Reservoirs (two – 30,000 gallon tanks, side by side) (MP 297.0)
 - f. Gillespie Gap Area – 10,000 Gallon Reservoir (MP 330.9)
 2. **Bid Additive 1 - locations:** (tank cleaning, and inspections)
 - a. Doughton Park Campground - 30,000 Gallon **Elevated** Reservoir (MP 239.2)
 - b. Bluffs Coffee Shop – 30,000 Gallon **Elevated** Reservoir (MP 241.1)
 - c.
 3. **Bid Additive 2 - locations:** (tank cleaning, and inspections)
 - a. Linville Area - 50,000 Gallon Reservoir (MP 316.4)
 4. **Bid Additive 3 - locations:** (tank cleaning, and inspections)
 - a. Crabtree Meadows Area - 10,000 Gallon Reservoir (MP 339.5)
 - b. Crabtree Meadows Area - 10,000 Gallon Reservoir (MP 339.5)
 - c. Crabtree Meadows Area - 30,000 Gallon Reservoir (MP 339.5)
- B. **Access:** Each reservoir project site may only be accessed from unimproved dirt roads which branch off from the paved motor road or parking lot. The closest access to some of the reservoirs is a 200-300 feet from a paved or unimproved surface.

1-4 CONTRACTOR-FURNISHED ITEMS

- A. The Contractor shall furnish all necessary supervision, personnel, equipment, transportation, materials, and incidentals required to complete the contract requirements.

1-5 WORK CONDITIONS AND LIMITATIONS

A. PROJECT OFFICERS AND INSPECTION

1. The Contracting Officer (CO) is the only person with authority to enter into, administer, modify and/or terminate the contract.

2. The Contracting Officer's Representative (COR) is a program official designated by the CO to assist in administering technical aspects of the contract. The COR cannot modify the contract.

B. FACILITY USE

The Plateau Districts have some the most popular visitor destinations on the Parkway with daytime and overnight recreational activities. The visitor services at all facilities are open daily from May through October, except for picnic areas which are open year-round.

C. REQUIRED PERFORMANCE PERIOD

1. Project Completed No later than December 30, 2026.
2. Submit project schedule for approval prior to mobilization.
3. Some scheduling may include working around different concessionaire operations to ensure their services do not impact the visiting public.

D. CONTRACTOR'S USE OF PREMISES

1. Confine on-site material storage, equipment and vehicle parking, and staging to areas specifically designated by the COR.
2. Keep job site clear of trash and debris. Dispose of any trash, debris and other unusable material legally at authorized facilities on non-Federal land.
3. On-site services such as electrical power, water, restrooms and telephones will be unavailable at the different project sites. Contractor shall make appropriate arrangements such as port-a-johns to accommodate their personnel.
4. Contractor shall make provisions to allow continued use of the general area outside the specific work zone.
5. Contractor shall comply with all applicable federal, state, and local requirements for protecting the safety of the Contractor's employees, park staff, and the environment. All applicable OSHA standards shall be followed when working on the project site.
6. Contractor shall submit two (2) copies of the written safety plan to the CO prior to starting the job. The Safety Plan shall cover, at a minimum, safety procedures, identified hazardous conditions, and emergency contact information for all personnel on-site.
7. Preservation of Natural Features: Confine all operations to work limits of the project. Prevent damage to natural surroundings. Restore damaged areas, repairing or replacing damaged trees and plants, at no additional expense to the Government.
 - a. Do not remove, injure, or destroy trees or other plants without prior approval.
 - b. Do not fasten ropes, cables, or guys to existing trees.

E. PARK-SPECIFIC CONDITIONS

1. Project-specific environmental compliance mitigations: The following items are required by the National Environmental Policy Act and shall be adhered to as applied to this project. Other sections of this statement of work may describe these requirements in more detail.
 - a. Approval for any chemicals used for cleaning shall be approved by Park Integrated Pest Management (IPM) Coordinator, Lillian McElrath. Allow 14 days for review and approval.
 - b. Staging for equipment and materials shall be as designated by the COR.
 - c. All chlorinated water, sludge, debris and other waste generated by the cleaning and disinfecting methods shall be properly disposed of in accordance with current guidelines and regulations.

1-6 QUALITY ASSURANCE

1. All materials and methods used for this contract must meet or exceed the current local, state, and federal codes, standards, and regulations applicable to the contract work which includes American Water Works Association (AWWA) standard C652 and Reference Manual 83A Public Health Drinking Water Protection and Prevention (Feb 16, 2018).
2. INDUSTRY STANDARDS/ CERTIFICATIONS
 - a. All work performed on this contract shall adhere to the most current standards of the following:
 - i. American Water Works Association (AWWA);
 - ii. RM-83A Public Health-Drinking Water-Protection and Prevention (NPS/PHS)
 - iii. U.S. Occupational Safety & Health Administration (OSHA);
 - iv. National Association of Corrosion Engineers International (NACE);
 - v. Society for Protective Coatings (SSPC);
 - vi. American Concrete Institute (ACI);
 - vii. Environmental Protection Agency (EPA); and,
 - viii. All State, local, and federal regulations related to the service performed.
3. All personnel performing work under this contract shall be trained and skilled in the trades required to complete the contract requirements. The Contractor shall meet the following minimum qualifications:
 - a. The Field Inspector shall have at least three (3) years of concrete water tank inspection experience.
4. The Contractor and/or subcontractor for this work shall demonstrate competence in water storage inspection cleaning and disinfection. All subcontractors shall be approved by the CO prior to starting work.
 - a. Contractor's firm shall be an established Potable Water Tank Cleaning company with a minimum of three (3) years' experience with the preparation of specifications for tank rehabilitation and completely familiar with the specific requirements and methods needed for proper performance of this Contract.

- b. Over the past three (3) years, Contractor shall have conducted a minimum of fifteen (15) inspections and cleanings of potable water tanks using those methods identified.
- c. Contractor's technical staff assigned to this Contract must have a minimum of one (1) year experience to perform any service under this contract.
- d. Contractor shall provide project references for at least five (5) recent projects and the resumes of the field inspector and report preparer.

1-7 REFERENCE ATTACHMENTS

<u>SOW Reference</u>	<u>Document Name</u>
Document	Water Reservoir Locations
Document	AWWA C652

PART 2: MATERIAL

2-1 GENERAL

- 1. All cleaning compounds must be Environmental Protection Agency (EPA), Food and Drug Administration (FDA), and National Sanitation Foundation (NSF) approved.
- 2. Contractor will use environmentally preferred products whenever available, e.g. Green Seal certified or equal.
- 3. Contractor will submit a list of the cleaning compounds, along with MSDS sheets, to the CO for review/approval prior to cleaning the tank(s).

PART 3: EXECUTION OF WORK

3-1 GENERAL

1. SERVICE INTERRUPTION

- a. There are no utility services to the facilities.

2. SITE INVESTIGATION

The Contractor is responsible for investigating site conditions in the vicinity of the required work to ensure accessibility for required equipment, vehicles and personnel or any other obstacles or items which may increase time or cost of the work.

3. SERVICE WORK

- a. Service shall be performed in a professional manner in accordance with industry standard best practices.
- b. All equipment introduced into the water shall be dedicated for potable water use only

and must be disinfected in accordance with AWWA C652 standard and local codes.

3-2 INSPECTION AND EVALUATION OF WATER STORAGE FACILITIES

1. Contractor shall perform a comprehensive evaluation of the interior and exterior conditions of the tank:
 - a. Inspection(s) shall be conducted with each tank full and in service and shall be completed without causing disruption to the treatment or distribution system.
 - b. No confined space entry will be permitted without the CO's or designee's approval and an approved safety plan.
 - c. All equipment used in the tank must be sprayed with a chlorine solution, at a concentration of 200-mg/L, in accordance with AWWA C652.
 - d. Contractor shall provide high quality color (DVD) video/digital recording inspection of 100% of all internal surfaces, including the roof, walls, floor, hatches, joints, gaskets, etc. through closed circuit TV. Video recording shall have real time imprint. Video shall be taken within a timeframe of very low turbidity so as to get clearest picture.
 - e. Contractor shall provide adequate lighting to thoroughly document all surfaces with video under all conditions of water and surfaces.
 - f. The Field Inspector shall be on-site during the inspection and shall witness and narrate the underwater camera video footage.
 - g. The Field Inspector shall provide, recorded on the inspection video, any professional assessment and evaluation comments about the condition of the tank.
 - h. Contractor shall be able during the inspection to remove coatings and corrosion products on the internal submerged surfaces for direct examination of metal loss, pit depth and/or coating failure.
 - i. Contractor shall review, evaluate and provide a written report of any issues identified in the inspection.
2. The following items shall be inspected on each tank:
 - a. Condition of concrete or steel on the interior including approximate percent of cracking, corrosion, settling, type of failure, and locations of concentrated spalling.
 - b. Corrosion of any exposed reinforcing materials.
 - c. Condition of any membrane covers, including flexibility and UV damage.
 - d. OSHA safety compliance of tank ladders and hatches.
 - e. Sanitary conditions of tank hatches, vents and other penetrations and interior plumbing.

- f. Percentage of coating failure and corrosion on any coated surfaces.
- g. Condition of foundation and anchors, visual inspection of obvious defects.
- h. Water level sensors and associated wiring.
- i. Seams, welds/bolted joints and integrity of previous repairs.
- j. All other structural members not mentioned above.
- k. Any other items required by the State of Virginia potable water tank inspection codes.

3-3 CLEANING OF WATER STORAGE FACILITIES

1. Tank Cleaning

- a. Contractor shall clean underwater surfaces to remove bio-growths, sediments, silt, sand, sludge buildup, etc.
- b. Contractor shall thoroughly remove all accumulated bottom sediment and debris from all potable water storage water reservoirs.
- c. All accumulated bottom sediment and debris shall be removed in a manner that does not compromise the tank integrity and/or coating system.
- d. Cleaning work shall not cause disruption to the use or quality of the water.
- e. The reservoir shall remain in operation while the cleaning is in progress. If shut down is necessary, it shall be for a minimal period.
- f. Cleaning of the tank floor will be performed with a vacuum pump and rotary bristle brush or equal.
- g. Cleaning shall remove the sediments while water system remains on-line.
- h. Turbidity shall be minimal, preserving water quality.

3-4 DISPOSAL OF WASTE MATERIALS

1. Disposal of Waste Materials

- a. Contractor will coordinate with the COR to designate the appropriate discharge point for waste and debris material collected.
- b. Sediment and water removed shall be pumped out of the structure to a COR designated discharge point.
- c. Contractor must meet all EPA and OSHA guidelines in the proper handling and disposal of waste.
- d. All debris removed from the tank(s) shall be disposed of in accordance with all

applicable federal, state, and local requirements, including any de-chlorination requirements.

- e. Neutralize all water-disposed from tanks being disinfected by treating with one of the chemicals listed in AWWA C652.

3-5 DIVER AND DIVING OPERATIONS

1. Any diving operations being proposed for this contract shall meet the following requirements:
 - a. All diving operations shall be conducted by certified commercial divers who have graduated from an Association of Commercial Diving Educators' Approved Commercial Diving Course or equal. Alternatively, divers who have completed specialized military training i.e., Navy 1ST or 2nd Class Dive Courses or Master Dive School shall also qualify.
 - b. All dive team members shall have a current CPR & First Aid Card, O2 Administrator Card, and a complete diver physical within the past 24 months.
 - c. Contractor shall provide certificates of training for all personnel conducting diving operations.
 - d. All diving operations shall be conducted in accordance with OSHA 29 CFR.
 - e. All diving operations shall be conducted utilizing a totally encapsulated diving dress, diver hardhat with sealed neck dam, and a vulcanized rubber dry suit, all in good repair. A band mask shall be specifically prohibited for any instance but in an emergency.
 - f. The diver hard hat shall be equipped with operating voice communication to the surface.
 - g. All diver equipment and any other equipment introduced into the reservoir shall be dedicated for potable water operations, and further, shall be disinfected with no less than a 200-ppm chlorine scrub/spray prior to entry into the reservoir.
 - h. Dive team shall be equipped with live color video with live voice recording (and associated lighting system) between diver and surface team, to allow for real-time surface monitoring of all diving activities and findings, as well as quality-control of the completed work.

3-6 ROV EQUIPMENT

1. Any Remotely Operated Vehicle (ROV) being proposed for this contract shall meet the following requirements:
 - a. Shall be dedicated for potable water use only and the Contractor shall provide a letter of certification stating the ROV is dedicated to potable water use inspection only.
 - b. ROV shall not contain any liquids that will contaminate the water. All thruster seal fluid

shall be a food grade product.

- c. Before placing ROV in potable water system, it shall be sanitized to meet AWWA Standard C652.
- d. Shall be able to abrade the underwater surfaces to clean off corrosion products and loose concrete.
- e. Shall be able to measure sediment depths on tank floor.
- f. Shall have a zoom camera capable of extreme close up inspection as small as 1/32" and capability for long distance inspection of above water surfaces within the tank.
- g. Shall be equipped with a vacuum pump and rotary bristle brush.
- h. Shall be equipped with a power wire cleaning tool that can clean off all surface debris, corrosion products and failed coating to permit the inspector to see pits, cracks and weld corrosion.
- i. Shall be equipped with adequate lighting for all conditions inside the tank.

3-7 PROJECT CLOSEOUT

A. CLEANUP

- 1. Remove all tools, equipment, excess materials, construction debris, and trash from the project site. Legally dispose of trash and debris at authorized facilities off Government property. Restore or refinish any surfaces damaged during performance of the contract work. Leave project site clean and ready for use at the time of final inspection.
- 2. Legally disposal of all excess and unusable materials at authorized disposal facilities.

B. FINAL INSPECTION AND ACCEPTANCE OF WORK

- 1. When all work including site restoration and cleanup has been completed, the Contractor shall contact the COR to arrange the final inspection. The final inspection should be conducted by the COR within 5 working days after receipt of the Contractor's request.
- 2. The Contractor will be notified of any deficiencies found during the final inspection. After all deficiencies have been corrected and all required closeout submittals have been forwarded to the COR, the CO will issue a letter of final acceptance and request the release of claims.
- 3. Contractor shall supply a "Certificate of Performance" for each reservoir to the CO at the completion of the project. At a minimum, the Certificate will indicate the date of cleaning, the name of the servicing company, areas cleaned, and those area(s) not cleaned.
- 4. Contractor shall submit four (4) copies of a detailed "Service Report" at the conclusion of the project to the CO (to include fully narrated DVD of the internal inspection). This Report shall indicate, but not be limited to, the following:

- a. Job site location and address
 - b. Date and time of cleaning
 - c. Type of tank
 - d. Name and certificate number of the technician(s)
 - e. Measurements and recording of bottom sediment depth
 - f. A high resolution, DVD video/digital recording providing a detailed review of structural concrete, steel, landings, etc. of the interior of the structure
 - g. Digital photos of ladders, piping (above ground), hatches, vents and all other structural/mechanical components
 - h. Recommended next service due date
 - i. Identify any mechanical, accessibility, and safety concerns
 - j. Identify any tank defects and recommendations for repair or upgrades
 - k. Identify in detail which component(s) is or/need repair/replacement/upgrade
 - l. Identify all repairs or upgrades needed to ensure the tank is in proper working order according to AWWA standards
 - m. A detailed cost estimate for any repair/upgrades found in the inspection/cleaning process.
5. Contractor shall provide a schematic diagram of each tank showing the location of each problem area with corresponding still color image.
 6. Contractor will provide detailed information and recommendations to maximize the serviceable life of each tank and provide information for possible modifications to improve operational effectiveness.

PART 4: MEASUREMENT AND PAYMENT

- A. Measurement and Payment: Payment for all labor, materials, equipment and services to complete the work in accordance to these specifications shall be included in the contract lump-sum price.
- B. Final payment shall be executed once the Government has formally accepted the work and provided the Letter of Acceptance to the Contractor.

END OF STATEMENT